

Ising Model using Metropolis Algorithm

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1 Introduction

Ising Model is a mathematical model in statistical mechanics and is used to describe the process of ferro-magnetism and para-magnetism. Usually, magnetic materials (at higher temperatures) have atoms arranged in a random manner, which leads to zero average magnetization. At a certain critical temperature T_c , phase transition happen inside the material, leading to alignment of the atoms in an ‘ordered’ fashion. The main aim of this project is to implement 2D Ising model numerically and observe the phase transition and the critical temperature.

1.1 Phase Transitions

A phase transition is a physical process of transition between states of the system. In the context of magnetic materials, an example could be ferromagnetic phase transition, from the state of being non magnetized to being magnetized, as one goes from a higher temperature to a lower temperature (lower than T_c). Phase transitions can broadly be of two types[5], namely First Order Phase transition, in which thermodynamic quantities derived from the first order derivatives of F(free energy) or G (Gibbs Free Energy) show a ‘Discontinuity’, or Second Order Phase transition, in which second order derivatives of F(or G) show a ‘singularity’. Ferromagnetic Phase transition is a second order phase transition, and hence we look at the behaviour of thermodynamic quantities like Heat Capacity(C_v) or Susceptibility(χ) in order to ascertain if and where the phase transition has happened.

1.2 2D Ising Model

Ising model describes these phenomena of phase transition in magnetic materials by using a discretized Lattice of ‘spins’ of the atoms, with each spin interacting with its nearest neighbours [3, 6]. In this description, one can look at the interaction Hamiltonian (2D) in order to extract the relevant physics.

$$H_{int} = -J \sum_{i=1, j=1, 2, 3, 4}^{N^2} S_i S_j \quad (1)$$

Here J is the coupling constant¹ and S_i is the spin of the i^{th} particle in the lattice and $S_{j=1,2,3,4}$ are the spins of nearest neighbours. Basically, for a particle at a lattice site (i, j) , its neighbours are situated at $(i + 1, j)$, $(i, j + 1)$, $(i - 1, j)$, $(i, j - 1)$ when $(i, j) \neq L$. Now suppose a spin at a random

¹Here we have assumed $J_{horizontal} = J_{vertical}$

↑	↑	↑	↑
↑	↑	↑	↑
↑	↑	↑	↑
↑	↑	↑	↑

Figure 1: 2D-Ising Lattice with all the spins in the same direction

site (i, j) gets flipped, then their would be change in energy of the entire configuration, which would be given by

$$dE = 2JS_{ij}[S_{i+1,j} + S_{i,j+1} + S_{i-1,j} + S_{i,j-1}] \quad (2)$$

We make use of the Canonical Ensemble to give the probability of a given microstate with energy E as

$$P(E)_T \propto e^{-E/K_B T} \quad (3)$$

In the absence of external field term, the 2D Ising model is analytically solvable and when we consider the Isotropic case as in eq. (1), then one gets a famous relation[1]

$$\frac{K_B T_c}{J} = \frac{2}{\ln(1 + \sqrt{2})} \simeq 2.27 \quad (4)$$

Although, the Ising model as described above is solvable theoretically in 2D, its not possible to say the same when we go to the 3D case or even a bit more complicated 2D case. Thus for general cases, we need numerical techniques. One of the famous and most commonly used algorithm to Implement 2D Ising Model is known as the ‘Metropolis Algorithm’.

1.3 Metropolis Scheme

Metropolis Algorithm [2] works by having a biased Monte Carlo sequence weighted by the Boltzmann weight $e^{-E/K_B T}$. To get the condition for Equilibrium, one can use the Master equation and then apply the principle of detailed balance as follows

$$\frac{dn_i}{dt} = \sum_k (P_{k \rightarrow i} n_k - P_{i \rightarrow k} n_i) \quad (5)$$

Here n_i is the number of particle in the i^{th} state and $P_{k \rightarrow i}$ is the probability of going from k^{th} state to the i^{th} state. If $\frac{dn_i}{dt} = 0$ then the RHS in eq. (5) should be zero. Then either the entire sum can be zero (this would be when their is a net current) or $(P_{k \rightarrow i} n_k - P_{i \rightarrow k} n_i) = 0$ for each k , which is the required condition for equilibrium. This gives us

$$\frac{P_{k \rightarrow i}}{P_{i \rightarrow k}} = \frac{n_i}{n_k} = e^{-\frac{E_i - E_k}{T}} \quad (6)$$

Now suppose $E_i > E_k$, and we set $P_{i \rightarrow k} = 1$, then

$$P_{k \rightarrow i} = e^{-\frac{E_i - E_k}{T}} \quad (7)$$

And this is exactly what we do while implementing the Metropolis Algorithm. When the spin flip lowers the energy, we accept that outcome with probability 1, and when it increases the energy, we accept it with $e^{-dE/T}$.

1.4 Thermodynamic Quantities

As discussed above in section 1.1, for the system in consideration, the type of phase transition is a second order phase transition. So in addition to looking at the behaviour of usual thermodynamic quantities like E and M , we would also look at C_V ² and χ , which are given as follows:

$$C_V = \frac{\partial E}{\partial T}|_B = \frac{\langle E^2 \rangle - \langle E \rangle^2}{K_B T^2} \quad (8)$$

$$\chi = \frac{\partial M}{\partial B}|_T = \frac{\langle M^2 \rangle - \langle M \rangle^2}{K_B T} \quad (9)$$

We expect a discontinuity in the above two quantities near the Transition temperature T_c , and thus this serves as a nice tool to check for the correctness of the implementation.

2 Algorithm

In order to use the Ising Model Hamiltonian and simulate it on a computer, we make use of the Monte-Carlo methods (Stochastic methods), which involves a random number generator, for accessing different microstates of the given lattice. One of the most common method to implement Ising Model is by using Metropolis Algorithm, and goes as follows:

- First, one needs initialize the lattice, by assigning spins at the each lattice point. This can either be done through random spin assignment at each point or by assigning all the sites a given spin. In the code implemented, we use random initialization. Also set up different temperature points at which one wants to calculate thermodynamic variables.
- After this, we choose a random lattice site and give that site a trial flip, and then calculate the change in energy of the configuration due to that flip. If the energy change is negative, then we accept that spin flip with probability 1. If the energy change is positive, then accept that spin flip with the probability $e^{-dE/K_B T}$. Do this for all the spins. This complete step of N^2 flips can be termed as one Monte Carlo move (mcmove).
- Repeat the above move many times, so that one can access all the spins sites (because the random distribution for choosing a lattice site is uniform), and thermalize the system (at a given temperature) with appropriate number of mcmoves and then, once the systems thermalizes, carry out additional mcmoves in order to get an ensemble average of thermodynamic quantities.
- Repeat the above steps for different temperature points to get thermodynamic averages at each temperature point. One can then plot a graph of Thermodynamic quantities w.r.t Temperature to observe the changes in order parameters and also get an idea about the critical temperature.

²One can derive this result as follows: $\langle E^2 \rangle - \langle E \rangle^2 = \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} - \frac{1}{Z^2} \frac{\partial Z}{\partial \beta}^2 = \frac{\partial}{\partial \beta} \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right) = -\frac{\partial \langle E \rangle}{\partial \beta}$, where Z is the Canonical partition function and $\beta = 1/K_B T$. Now $-\frac{\partial \langle E \rangle}{\partial \beta} = K_B T^2 \frac{\partial \langle E \rangle}{\partial T} = K_B T^2 C_V$

3 Numerical Implementation

In this section, we will discuss about some specific steps which were followed for implementing the algorithm. While we need to know the theory and algorithm for the numerical implementation of any given model, one also needs to take into consideration that computers don't understand the theory in the same way as we do. In this context, they do not understand the units and thus, we need to compute everything in a dimensionless manner, so that the computers can do the relevant numerical computations and then we can later add the units and make sense of the results.

We start with defining the values of Spins as either +1 or -1 (instead in terms of $\hbar$). Then we identify Energy and thus temperature in terms of the coupling constant J . Since $[E] = [J] = [K_B T]$, we can write everything in terms of J^3 . In the code, we take both J and K_B equal to 1, which we can re-scale later. In this way, we will actually set the temperatures in the units of $\frac{J}{K_B}$.

Now once the units are in the place, we need to set up the periodic boundary conditions. This is because, we are using a lattice, and that too a small one, as compared to the real material (where the surface atoms make up only a small fraction of the atoms), we need to take into account periodic boundary conditions in order for the simulation to mimic real phenomena. The boundary conditions are set such that when we are looking at the lattice points at the boundaries, then their neighbours are the one at the opposite boundary. For ex, consider a Lattice of size $L \times L$, then for if we consider a spin at the rightmost column (in a given row) in the lattice, then its right neighbour is set as the leftmost spin in the same row. In the above case if the position of the particle is at (i, L) , then its neighbours lie at $(i + 1, L)$, $(i, 1)$, $(i - 1, L)$, $(i, L - 1)$, and so on.

After setting up the boundary conditions, we initialize the lattice and then we can carry out the thermalization (evolving the system to thermal equilibrium). For carrying out the metropolis scheme, we use the fact that the probability of a flip with positive energy difference is given by $e^{-dE/T}$ and to implement this condition we call a random number (u) with uniform probability distribution between 0 and 1. If $u < e^{-dE/T}$, then we accept the trial flip and update the energy and magnetization values. We continue this Monte Carlo step sufficient number of times so that the system reaches the thermal equilibrium. After this we carry out further Monte Carlo steps in order to take average of the thermodynamic quantities. The entire code for this has been implemented in Python⁴ and is based on [4] with major modifications. Although Python is not the most ideal languages for simulations like these (its painfully slow)⁵, it is still one of the easiest languages to understand and implement numerical algorithms.

4 Artifacts of Simulation

Given the simplicity of the 2D Ising Model, we can run into numerous issues if we are not careful about some of the caveats of the numerical implementations. Since we are working with finite size lattices, that too small lattices owing to limited available computational power, we need to carefully look at the various factors which affect the final results.

Lattice Size L is a very important parameter in the Ising 2D model, and choosing this appropriately determines the accuracy of the results. Lower L values lead to more relative uncertainty when we look at the graphs of E, M, C_V, χ vs T . For low values of L , say $L = 10$, the magnetization

³An example for this is, $K_B T$ at $T = 300K = 3.14 \times 10^{-21} \text{ Joules}$ and $1eV = 1.6 \times 10^{-19} \text{ Joules}$ which implies $1eV \simeq 0.38 \times 10^2 K_B T_{300}$. If we assume $J = 0.05eV$, then $n \times J = n \times 0.05eV$. Using this, we can write $K_B T$ in terms of J . For example, if $K_B T = 0.1eV$ then this implies $K_B T = 2J$, since $J = 0.05eV$

⁴The link to the the code: Jupyter Notebook

⁵Numba decorator has been used to speed up the code and this results in decreasing the computation time by a factor ~ 30

of the lattice can flip easily for $T < T_c$, and would naively give 0 average magnetization. This is because for lower L values, the amount of energy needed to flip is not that large, when compared to large L values ($E \sim N^2$). One way to deal with this, which has also been implemented in the code is to use the absolute value of M . While it does solve the issue of the flip of the magnetization values below T_c , it does so at the expense of a little higher average magnetization at higher temperatures, which can be reduced by taking larger and larger values of L . This is evident in fig. 2b, where larger L values converge closer to 0 average magnetization for higher temperatures.

Value of L also dictates what the **relaxation time** τ (number of steps needed to thermalize the system) and **number of mcmoves** should be for averaging of the thermodynamic quantities. For example, for a lower value of L , the relaxation time at a given temperature would be less, when compared to large L . Also, the number of mcmoves required for averaging thermodynamic quantities would be much more for larger values of L , because they have more number of overall spins ($\tau \sim L^2$ for 2D). In case of relaxation time at a given temperature, it is useful to note that at lower temperatures, the system takes more time to reach equilibrium as compared to higher temperatures.

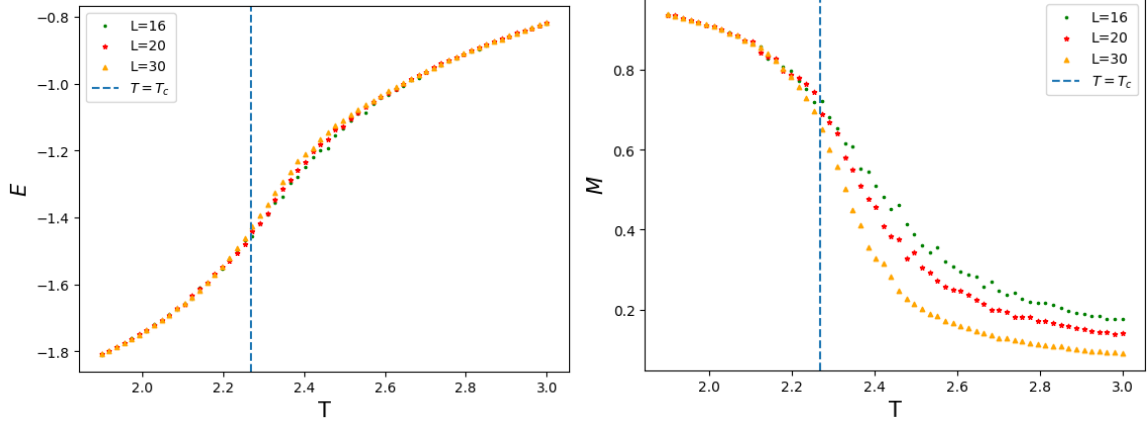
One can also see that upon increasing the value of L , the value of peak of the C_V and $\mathcal{X}$ vs T graphs come closer to the actual value of T_c (2.27), and the peaks themselves become more sharp, and this can be clearly observed in fig. 3a and fig. 3b.

Thus when running the simulation, we need to take appropriate combinations of these parameters in-order to get correct results. Some of the consequences of the above discussion can be clearly seen in the Results section 5.

5 Some Preliminary Results

In this section, presented are some of the results of the numerical simulations performed. For different value of lattice size L , different values of relaxation time(In this case, number of Monte Carlo steps to equilibrate) and number of mcmoves have been used, in-order to reduce computational times. But sufficient number of steps have been given for all the cases to have enough time to thermalize and then take the ensemble averages. The relaxation time is the same (10,000 steps) for all the cases included here in figs. 2 and 3. The number of mcmoves range between 10^5 to 2.5×10^5 . For all the calculations, absolute value of magnetization has been used. Value of critical temperature $T = T_c = 2.27$ as in eq. (4) has been plotted for comparisons of the accuracy of the simulations.

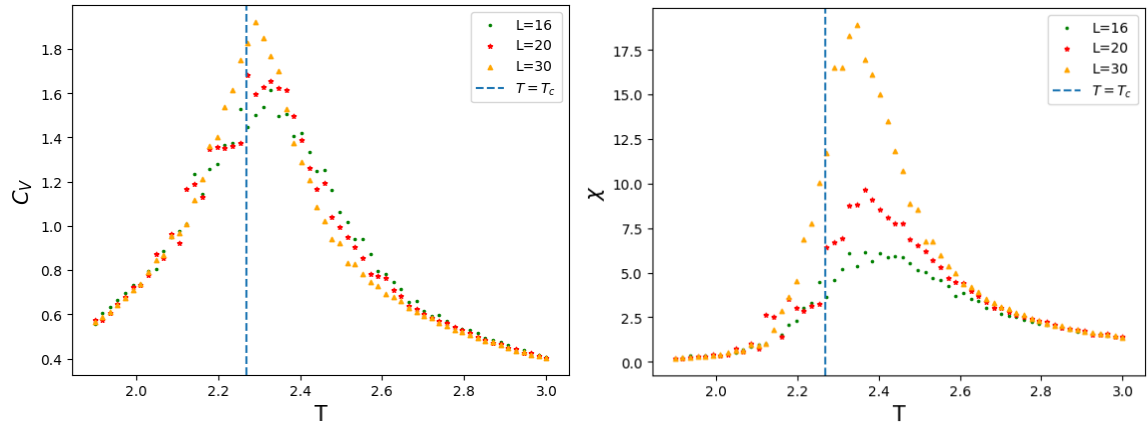
In the fig. 2 one can observe the phase transition happening as we go from higher temperatures to below T_c , the energy value converges to a lower value and the magnetization per particle also goes to 1, indicating an ordered phase. And in fig. 3 one can observe the discontinuity which peaks near T_c , thus validating the theoretical prediction.



(a) Energy per particle vs Temperature $\frac{J}{K_B}$

(b) Magnetization per particle vs Temperature $\frac{J}{K_B}$

Figure 2: Plots of first order derivatives (E and M) of Free energy vs Temperature.



(a) Heat Capacity C_V vs Temperature $\frac{J}{K_B}$

(b) Magnetic Susceptibility χ vs Temperature $\frac{J}{K_B}$

Figure 3: Plots of second order derivatives (C_V and χ) of Free energy vs Temperature.

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